Restaurant Sales Analysis

Power BI project report: dataset, preparation, calculations, insights and recommendations

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Tools: Power BI, Power Query, DAX

Total Sales	Total Orders	Items Sold	Avg Order Value
299,071	500	2,326	598.14
Avg Delivery Time	**Avg Food Rating**	**Avg Delivery Rating**	**Customers**
30.54 mins	3.36	2.99	21

1. Project overview

This project analyzes restaurant order data to answer practical business questions: when do customers order, which cuisines and zones perform best, how do customers pay, and how good is the delivery experience? The data was cleaned and modelled in Power BI (Power Query for preparation, DAX for measures) and presented in a single-page sales dashboard. It was inspired by a published Excel and Power BI restaurant-sales walkthrough, but the model, measures and conclusions here are built independently, with some corrections noted in Section 7.

2. The dataset

Two Excel files are used. They connect through the restaurant ID.

Table	Rows	Columns
Orders Data (Orders.xlsx)	500	Order ID, Customer Name, Restaurant ID, Order Date (date and time), Quantity of Items, Order Amount, Payment Mode, Delivery Time Taken (mins), Customer Rating-Food, Customer Rating-Delivery
Restaurant Details (Restaurants.xlsx)	20	RestaurantID, RestaurantName, Cuisine, Zone, Category (Pro or Ordinary)

2.1 Data quality checks

Check	Result
Blank values	None in any column
Duplicate Order IDs	None (500 unique orders)
Orders with no matching restaurant	0 (every Restaurant ID matches)
Date range	Single day: 01 Jan 2022 11:10 AM to 11:58 PM
Quantity of items	1 to 7 per order
Order amount	3 to 1198; 16 orders are below 50 and may be worth checking, but they were kept
Delivery time	10 to 50 minutes
Ratings	1 to 5 for both food and delivery
Customers and restaurants	21 customers, 20 restaurants (4 zones, 8 cuisines)

Important limitation: all orders are from one day, so the analysis shows patterns within a day. It cannot show weekly or monthly trends, and no Date table was needed.

3. Data preparation (Power Query)

Step	What was done	Why
1	Loaded both Excel files and set data types	Correct types for dates, numbers and text
2	Renamed Customer Rating-Food and Customer Rating-Delivery to Food Rating and Delivery Rating	Cleaner names that are easier to use in DAX
3	Added an Hour column from Order Date (Add Column, Time, Hour)	A plain number is easier and safer to compare than a full date-time
4	Created Time of Day with a conditional column on Hour	Groups orders into four periods for analysis
5	Created Payment Type: Cash on Delivery = Cash, everything else = Card	Simplifies three payment modes into two
6	Renamed RestaurantID to Restaurant ID in the Restaurants table	Both key columns now share a name
7	Close and Apply	Loads the cleaned tables into the model

Time of Day cutoffs used in the final dashboard (based on the Hour column):

Time of Day	Hours	Orders
Morning	before 12:00	73
Afternoon	12:00 to 16:59	217
Evening	17:00 to 18:59	38
Night	19:00 onward	172

4. Data model

Instead of merging everything into one flat table, the two tables are kept separate and related in the model (a simple star schema). The relationship is **Restaurant Details[Restaurant ID] (one) to Orders Data[Restaurant ID] (many)**, active, single-direction filtering. One restaurant has many orders. This keeps restaurant attributes (cuisine, zone, category) in one place and lets any visual slice orders by them.

5. Calculations (DAX measures)

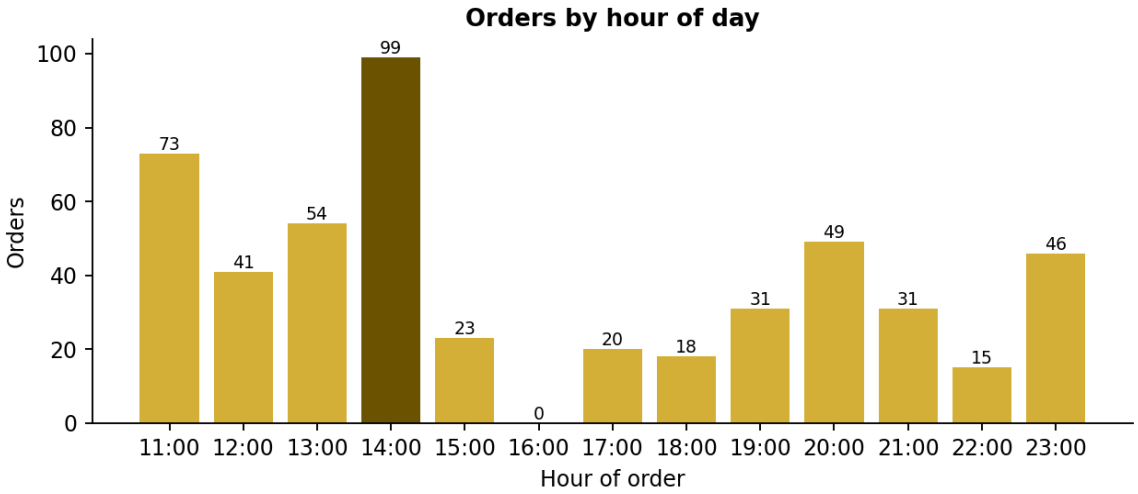
Measure	DAX formula	Result
Total Sales	`Total Sales = SUM('Orders Data'[Order Amount])`	299,071 (shown as 299K)
Total Orders	`Total Orders = COUNTROWS('Orders Data')`	500
Items Sold	`Items Sold = SUM('Orders Data'[Quantity of Items])`	2,326
Avg Delivery Time	`Avg Delivery Time = AVERAGE('Orders Data'[Delivery Time Taken (mins)])`	30.54 mins
Avg Food Rating	`Avg Food Rating = AVERAGE('Orders Data'[Food Rating])`	3.36
Avg Order Value	`Avg Order Value = DIVIDE([Total Sales], [Total Orders])`	598.14
Avg Delivery Rating	`Avg Delivery Rating = AVERAGE('Orders Data'[Delivery Rating])`	2.99

Why these are separate: COUNTROWS counts orders (500) while SUM of Order Amount totals money (299K). Some published versions of this analysis label the sales amount as "Total Orders", which mixes the two. DIVIDE is used instead of the / operator because it returns blank rather than an error if the denominator is zero.

Customer count on the dashboard is a distinct count of Customer Name, which gives 21.

6. Analysis and findings

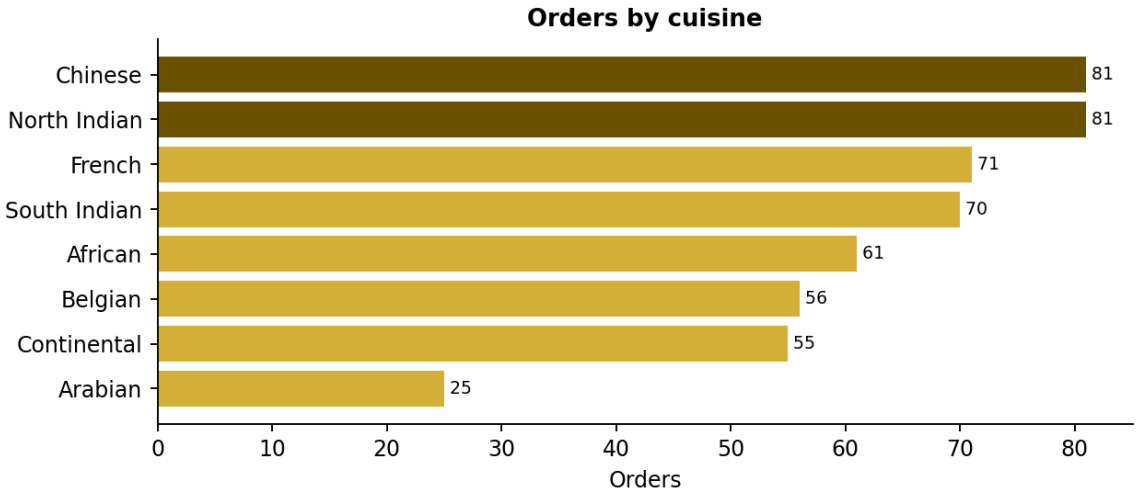
6.1 Time of day



Time of Day	Orders	% of orders	Sales	Avg order value	Avg delivery (mins)
Morning	73	14.6%	47,296	647.9	29.4
Afternoon	217	43.4%	127,074	585.6	29.9
Evening	38	7.6%	21,280	560.0	32.9
Night	172	34.4%	103,421	601.3	31.3

Afternoon is the busiest period with 43% of all orders. The peak hour is 2 PM with 99 orders, about 1 in 5 of the day. There are no orders at 4 PM, and demand rises again from 7 PM.

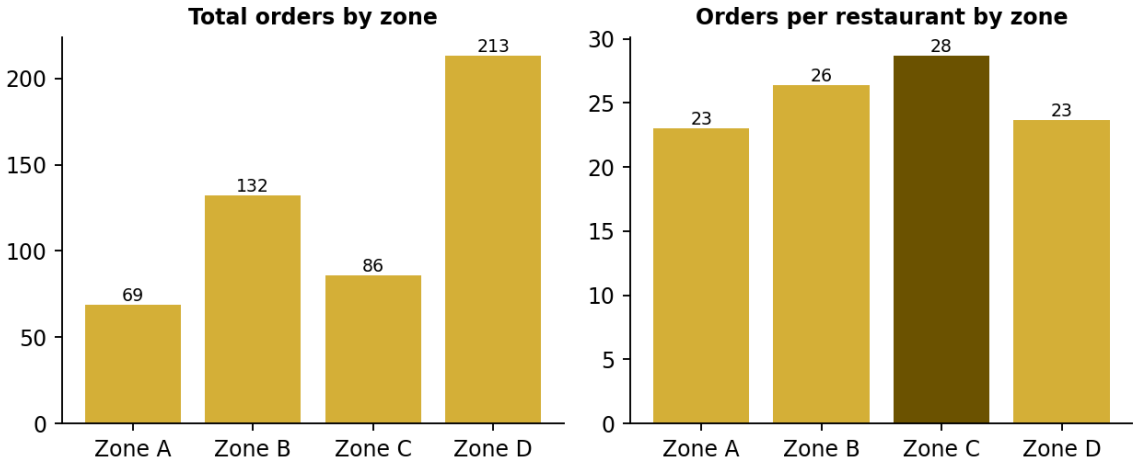
6.2 Cuisine



Cuisine	Orders	Sales	Avg order value	Food rating	Delivery rating
Chinese	81	45,437	561.0	3.31	2.85
North Indian	81	46,706	576.6	3.59	3.00
French	71	45,121	635.5	3.27	2.96
South Indian	70	38,855	555.1	3.21	3.29
African	61	36,452	597.6	3.46	3.08
Belgian	56	34,996	624.9	3.25	2.64
Continental	55	34,882	634.2	3.38	3.07
Arabian	25	16,622	664.9	3.44	3.08

Chinese and North Indian tie for the most orders (81 each). North Indian also has the best food rating (3.59). Belgian has the weakest delivery rating (2.64). South Indian has the lowest average order value (555) but the best delivery rating (3.29).

6.3 Zone



Zone	Restaurants	Orders	Orders per restaurant	Sales	Avg order value
Zone A	3	69	23.0	40,833	591.8
Zone B	5	132	26.4	77,001	583.3
Zone C	3	86	28.7	53,074	617.1
Zone D	9	213	23.7	128,163	601.7

Zone D has the most orders (213, 43% of all orders), but it also has 9 of the 20 restaurants. Per restaurant, Zone C performs best (28.7 orders each, against 23.7 in Zone D). Raw totals therefore overstate Zone D's advantage.

6.4 Payment, top restaurants and top customers

Payment type	Orders	% of orders	Sales	% of sales
Card	331	66.2%	200,917	67.18%
Cash	169	33.8%	98,154	32.82%

Top 5 restaurants by sales	Sales	Orders	Food rating	Delivery rating
Veer Restaurant	19,168	29	3.10	2.69
The Cave Hotel	18,934	32	3.59	3.50
Anand Restaurant	18,589	29	3.38	3.07
Willies	18,324	30	3.27	3.03
Ellora	17,863	32	3.53	3.09

Top 5 customers by sales	Sales	Orders
Srini	20,345	34
Revandh	18,006	30
Veer	17,989	26
Rifa	17,458	28
Ram	16,854	29

6.5 Ratings and delivery

The average food rating is 3.36 but the average delivery rating is only 2.99, even though delivery takes about 30.5 minutes on average. Delivery is the weakest part of the customer experience.

Restaurant category	Orders	Avg order value	Food rating	Delivery rating
Ordinary	321	592.9	3.37	2.98
Pro	179	607.6	3.34	3.02

7. Key insights

• **Lunch is the rush.** Afternoon brings 217 of 500 orders (43%), and 2 PM alone has 99. Orders drop to zero at 4 PM, then recover at night (172 orders).

• **Zone D leads on total orders, but that is misleading.** It has 9 of 20 restaurants. Zone C does best per restaurant.

• **Chinese and North Indian are the most ordered cuisines** (81 each). North Indian has the best food rating.

• **Delivery is the weak spot.** Delivery rating (2.99) trails food rating (3.36). Belgian is lowest at 2.64.

• **Card payments dominate:** 66.2% of orders and 67.2% of sales.

• **A small, loyal customer base:** 21 customers placed all 500 orders, and the top customer placed 34.

8. Recommendations

• **Staff for the lunch peak.** Add delivery and kitchen capacity around 12:00 to 15:00, with extra attention at 2 PM.

• **Look at the 4 PM gap.** A mid-afternoon promotion could test whether demand can be moved into that dead hour.

• **Improve delivery quality.** Investigate low delivery ratings, starting with Belgian restaurants, since food ratings are healthier.

• **Choose locations by per-restaurant performance,** not total orders. Zone C looks stronger than its total suggests, so Zone D may already be crowded.

• **Protect and grow the customer base.** With 21 customers driving all orders, loyalty offers and acquiring new customers both matter.

9. Corrections to the reference article and dashboard notes

• The article labels 299K as "Total Orders". It is the total order amount. The number of orders is 500, and 2,326 is items sold.

• The article's Time of Day screenshot shows some mislabelled periods (for example an 11:15 AM order marked as Night). Here the buckets are built from a clean Hour column.

• The article recommends opening in Zone D based on total orders. Adjusting for the number of restaurants changes that conclusion.

• Dashboard polish: fix the title spelling (RESTAURANT), set the Avg Food Rating gauge maximum to 5 (it currently ends at 6.72), and change "Number of customer" to "Number of Customers".

10. Limitations

The dataset covers one day (1 January 2022), so results describe that day only and may not generalize. Order Amount has no stated currency. There are only 21 customers, so customer-level patterns are based on a small group. Sixteen orders have amounts below 50, which could be genuine small orders or data entry issues.

Appendix: LinkedIn post draft

I analyzed 500 restaurant orders in Power BI, and one insight surprised me.

I built a Restaurant Sales Dashboard from scratch, using Power Query for cleaning, a data model linking orders to restaurants, and DAX measures for the KPIs.

What the data showed:

- Lunch is the rush: 43% of all orders came in the afternoon, with a single peak hour (2 PM) driving about 1 in 5 orders
- Chinese and North Indian cuisines lead with 81 orders each
- Two-thirds of orders are paid by card
- Delivery rating (2.99) lags food rating (3.36), so the delivery experience has the most room to improve

The surprise: Zone D has the most orders overall, but it also has 9 of the 20 restaurants. Per restaurant, Zone C performs best. Raw totals can point you to the wrong conclusion.

One caveat: this dataset covers a single day of orders, so it shows patterns within a day rather than long-term trends.

Tools: Power BI, Power Query, DAX

This was a great exercise in going beyond "which number is biggest" to asking "is that number fair?"

I'd love your feedback on the dashboard. What would you add?

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